When the Dog Has Inertia: Exact Feedback Equilibria and Equilibrium Selection in a Potential Influence Game

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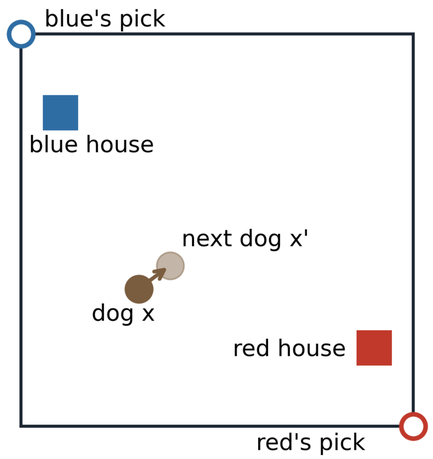
Code, data, full report: https://charith-reddy-pareddy.github.io/dog-game/

1. The dog game

Two players each pick a point in the unit square. A shared dog moves part of the way toward their weighted average, and each player wants the dog near its own house:

$$x' = w0 \cdot x + (1 - w0) \cdot (w \cdot a_{red} + (1 - w) \cdot a_{blue})$$
$$\text{reward}_i = -\|x' - h_i\|^2, \text{ discount } \gamma = 0.9$$

The dog's own weight w0 is its **inertia**; w0 = 0 is the one-shot game.



2. What was known

With no inertia this is an instance of the Battling Influencers Game (Wu et al., 2025), an **exact potential game**: with $d = w \cdot a_{red} + (1 - w) \cdot a_{blue}$,

$$\Phi = -\|d\|^2 + 2w \cdot h_{red} \cdot a_{red} + 2(1 - w) \cdot h_{blue} \cdot a_{blue}$$

changes by exactly a deviating player's reward change, so equilibria maximize Φ. In a coordinate the two houses share, Φ is flat: a continuum of equilibria.

3. Questions

- With inertia, is the one-shot equilibrium still a **feedback (Markov-perfect) equilibrium**, and if not, what is?
- Is the dynamic game still a potential game?
- On a continuum of equilibria, which one does **policy-gradient self-play** find, and why?

4. Methods

- **Exact:** the game splits by coordinate; each is solved by Nash value iteration (401 dog positions × 101 actions) and checked with the one-shot deviation principle using closed-form values.
- **Learning:** REINFORCE self-play with a Gaussian policy (sigmoid mean, clipped actions), 1500 episodes, 50 seeds per setting, plus two ablations.
- Open source and reproducible with an automated test suite; every number on this poster is generated from the saved results.

9. Known and new

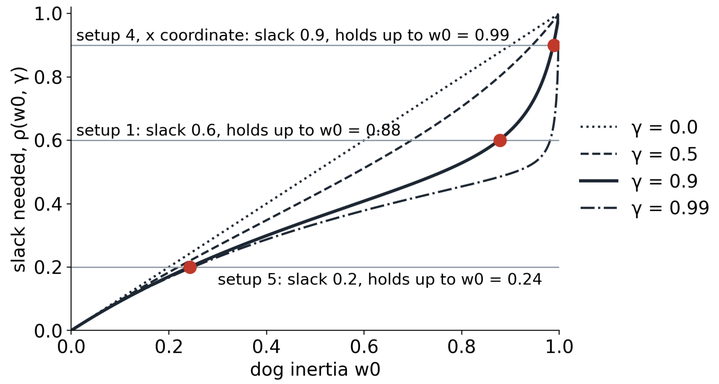
- **Known:** the one-shot potential structure (Wu et al., 2025); a game of potential stage games need not be a Markov potential game (Leonardos et al., 2022).
- **New for this game**, as far as the literature checked goes: Propositions 1 and 2, the Markov-potential counterexample, the selection mechanism with ablations, and that self-play misses the feedback equilibrium.
- **Limits:** the proofs (in the repository) are not yet refereed; the interior case is open; one learning algorithm on one game.

Wu, Zhu, Cai, Zhu (2025), Battling Influencers Game, arXiv 2502.01127 · Leonardos et al. (ICLR 2022), Markov potential games · Monderer and Shapley (1996), Potential games · Fudenberg and Tirole (1991), Game Theory.

5. When does the one-shot equilibrium survive inertia?

Let $k = (1 - \gamma \cdot w0^2)/(1 - \gamma \cdot w0)$ and give each player the target $T = d^* + k \cdot (h - d^*)$, where d^* is the one-shot Nash dog.

Proposition 1. Holding the one-shot actions at every dog position is a feedback equilibrium if and only if every player at an edge of the box has slack $s \geq \rho(w0, \gamma) = w0(1 - \gamma w0)/(1 - \gamma w0^2)$, where $s = (h - d^*)/(1 - d^*)$ at the top edge, and no one-shot action is interior.



Holding is an equilibrium while the curve (γ = 0.9 in bold) stays below a setup's slack; the dots mark where it stops. More patient players tolerate more inertia.

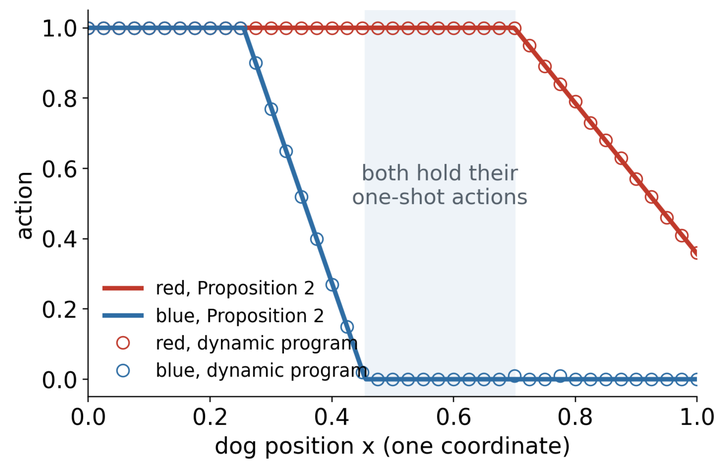
- Predicted vs computed share of dog positions where holding fails: within about one percentage point in all 30 coordinate games of the five setups.

6. What players do instead

Proposition 2. When the houses lie on either side of the dog, the policy in which each player aims the next dog position at its own target, as far as its weight allows,

$$a_i(x) = \text{clip}(\, a_i^* + (T_i - w0 \cdot x - (1 - w0) \cdot d^*) \,/\, ((1 - w0) \cdot w_i),\, 0,\, 1\,)$$

is a feedback equilibrium. It is piecewise affine, and equals the one-shot profile wherever holding is optimal. *Proof idea:* each player's one-round objective is single-peaked at its target.



Setup 3, x coordinate, w0 = 0.6: the explicit policy (lines) and the dynamic program (circles).

- Within one action-grid step (0.01) of the dynamic program in every such coordinate; one-round deviation gains below 1e-31.
- 200 random games: an equilibrium in 77 of 77 straddling games; it fails in 53 of 123 games with an interior one-shot action, which stays open.
- **Learners miss it.** Even starting every episode off the path, A2C and PPO self-play learn to hold the one-shot actions everywhere, exploitable off the path, even with 5x longer training or a Beta policy.

7. Not a Markov potential game

In a Markov potential game every 4-cycle of unilateral policy changes sums to zero. Exact values, setup 1:

w0	policies ignore the dog	policies react to the dog
0.0	+6.7e-15	+7.90
0.3	+3.1e-15	+7.12
0.6	+1.8e-15	+5.63
0.9	+5.6e-15	+1.85

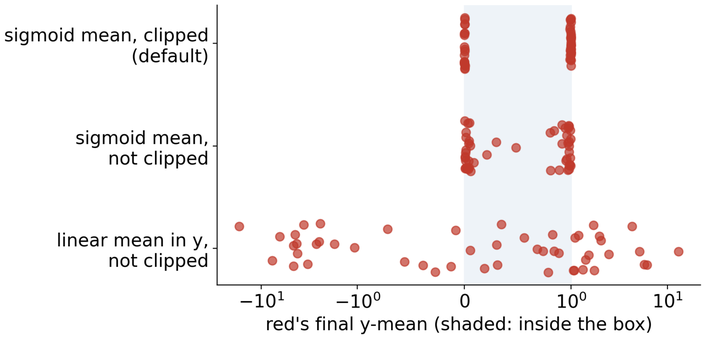
- Open-loop it stays a potential game at any inertia; with feedback it does not, even at w0 = 0. So convergence guarantees for independent policy gradient in Markov potential games (Leonardos et al., 2022) do not apply.

8. Selection on a flat potential is a random walk

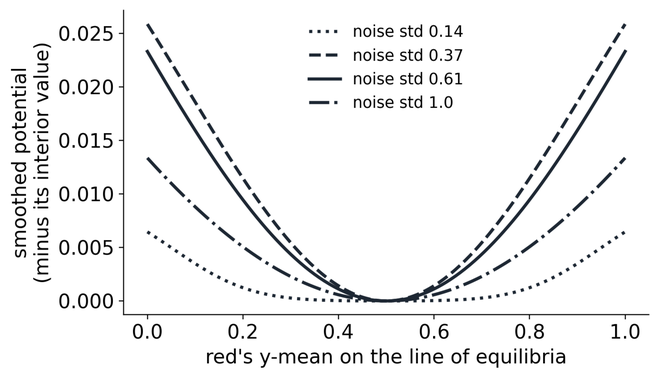
Noise std	red top	red bottom	interior	p (fair coin)
0.14	19	22	0	0.76
0.37	31	19	0	0.12
0.61*	31	19	0	0.12
1.00	30	20	0	0.20

50 seeds per noise level, shared-y setup. *Default.

- **Never the interior equilibrium** (0 of 250 runs). **Which mirror image: a fair coin.** An earlier 10-seed effect of the noise did not replicate.



- The expected gradient along the flat direction is zero but noisy, so the means random-walk; the box stops them where the sigmoid saturates. With an unbounded mean the walk leaves the box (38 of 50 runs).



- With clipped exploration noise, the averaged potential even dips at the interior: a push to either edge, equally.

Take-home

- With inertia the one-shot equilibrium survives on the path, and everywhere exactly when each player's slack is at least ρ(w0, γ); otherwise players aim the dog at an explicit target.
- The static potential does not carry over to feedback play.
- On a flat potential, learning is a random walk that the box stops; the mirror image is a coin flip.